



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
PRE-SCORE
15-10-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

PAPER-1

Time : 3 Hours

Maximum Marks : 216

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

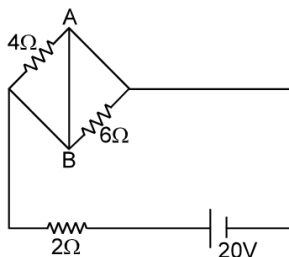
Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

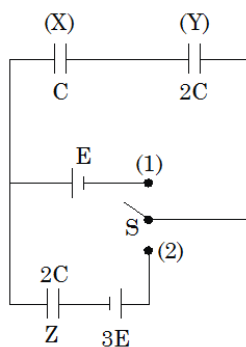
1. Because of the effects of air drag abandoned satellites, at the end of their usual lives lose energy in the upper layers of atmosphere, before finally burning up when they reach the denser lower layers. It can be shown that satellites originally moving along circular trajectories will continue to travel in approximately circular orbits, with their orbital radii slowly decreasing. Drag force acting on a satellite can be expressed as $F_{\text{drag}} = C\rho v^2$ where ρ is density of air at the altitude of satellite and v is the speed of satellite, C is a constant. Mark the correct statement(s) :-
 - (A) During slow decrease in radius of orbit, gravitational force is not perpendicular to velocity of satellite.
 - (B) Speed of satellite decreases as it approaches earth due to tangential drag force.
 - (C) If change in orbital radius of Δr during a single revolution then expression for density of air is $\rho = \frac{1}{4\pi C} \frac{m}{r^2} \Delta r$
 - (D) If change in orbital radius of Δr during a single revolution then expression for density of air is $\rho = \frac{1}{2\pi C} \frac{m}{r^2} \Delta r$

Space for Rough Work

2. In the circuit shown in the figure



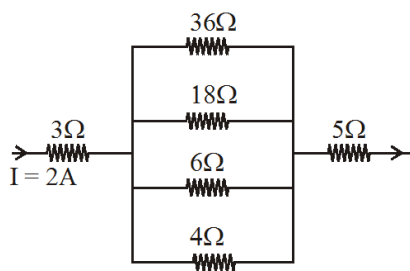
- (A) power supplied by the battery is 200 watt.
 (B) current flowing in the circuit is $(22/5)$ A.
 (C) potential difference across 4Ω resistance is equal to the potential difference across 6Ω resistance.
 (D) current in wire AB is zero.
3. In the given circuit diagram initially all the capacitors are uncharged. Now switch S is closed with position (1) and when steady state reached switch is shifted to position (2) from (1).



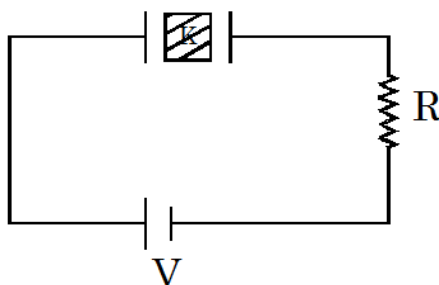
- (A) Final charge on all three capacitors will be same.
 (B) Final charge on capacitor Y is $\frac{4CE}{3}$.
 (C) Total heat developed in the circuit after shifting the switch from position (1) to position (2) is $4CE^2$.
 (D) Total heat developed in circuit from starting is $\frac{26}{3}CE^2$.

Space for Rough Work

4. Current passing through 3Ω resistor, as shown in diagram is $2A$. Then which of the following statements are correct ?



- (A) Current passing through 4Ω is $1A$.
 (B) Potential difference across 6Ω resistor is 4volt .
 (C) Power consumption in 18Ω is $\frac{8}{9}\text{watt}$.
 (D) Current passing through 36Ω is $1A$.
5. Consider the circuit shown in diagram. Capacitor is fully charged. Now dielectric slab is suddenly removed by an external agent. Equilibrium is again reached. During this time



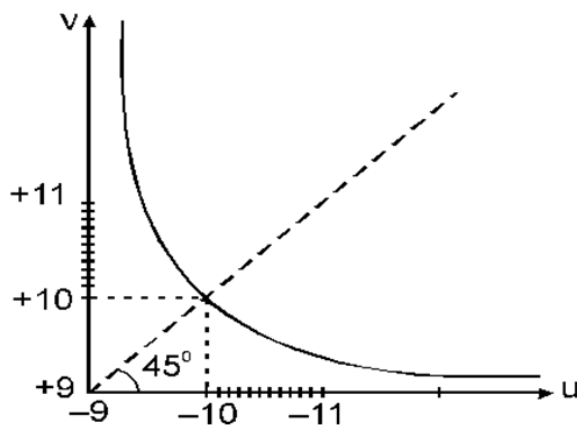
- (A) Work is done on battery.
 (B) Heat is produced in resistance.
 (C) Work is done by battery.
 (D) Loss in energy of capacitor is equal to energy given to battery and heat produced in resistance R .

Space for Rough Work

6. Choose the correct statement(s) :
- (A) The electric potential at a point on the axis of an electric dipole depends on the distance r of the point from the dipole as $\frac{1}{r^3}$.
 - (B) Electric dipole moment is a vector quantity.
 - (C) When a test charge is brought from infinity to any point along the perpendicular bisector of the electric dipole the work done is zero.
 - (D) An electric dipole kept in a uniform electric field may experience torque.
7. Consider a capacitor of capacitance $10\mu\text{F}$ connected to a battery. The space between the parallel plates is filled slowly with dielectric of dielectric constant 5. Choose the correct option(s) :
- (A) If the capacitor was connected to a battery throughout the process electric field before and after insertion of dielectric remains same.
 - (B) If the capacitor was disconnected from the battery & then dielectric was inserted, electric field before & after the insertion of dielectric remains same.
 - (C) Break down potential of capacitor increases after insertion of dielectric.
 - (D) Capacitance of the capacitor increases.

Space for Rough Work

8. The graph between object distance u and image distance v for a lens is given below. The value of u and v are in centimeter. Then mark the correct statement(s).



- (A) The focal length of the lens is 5cm.
 (B) The focal length of the lens is 0.5cm.
 (C) The error in focal length is 0.05 cm.
 (D) The fractional error in focal length is 0.1.

Space for Rough Work

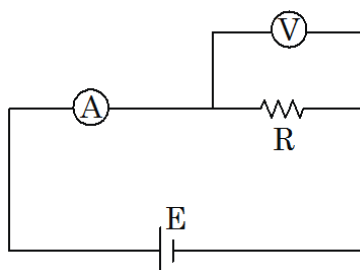
SECTION-II (i) : (Maximum Marks: 8)

- This section contains **FOUR (04)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A binary star is a system of two stars moving around the centre of inertia of the system due to mutual gravitational force. If the total mass of two stars = M_0 and the period of revolution is T . Find the distance in km between stars ? (Take $GM_0 = 8 \times 10^6$ and $T = 2\pi \times 10^3$ seconds)
2. A voltmeter and an ammeter are connected in the circuit as shown. Resistance of ammeter is say $\frac{R}{10}$ and that of voltmeter is $10R$. Percentage error in the reading of voltmeter (compared to the case when meters are ideal) is $-x\%$ then find the value of $\frac{x}{5}$ (approx.).



Space for Rough Work

3. What is the force/length between two long parallel current carrying wires carrying current of 186 A and 24 A at a distance of 0.1 mm from each other. Give the answer correct to significant digits (in N/m).
4. A point charge is kept on the axis of a circular non conducting disc. The distance is such that $\frac{1}{4}$ th of the total electric flux coming from the charge passes through the disc. By what distance (in cm) should the charge be moved along the axis so that $\frac{1}{5}$ th of the total electric flux coming from the charge passes through the disc ? The radius of disc is 30 cm.

Space for Rough Work

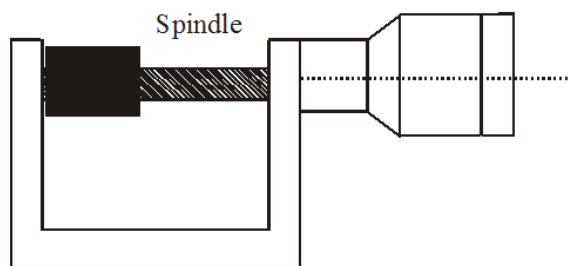
SECTION-II (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

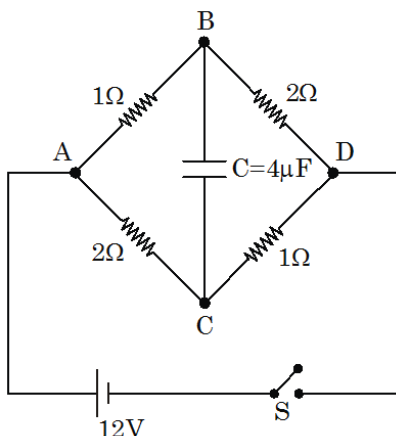
Zero Marks : 0 In all other cases.

5. In the diagram shown, 12 complete threads of the spindle are visible from every angle. 56th csd coincides with the reference line. Initially, 35 complete threads were present and visible on the spindle. Initially means where there was no object between the stud and screw. 1msd = 100csd = 1mm. Length of the body is mm.

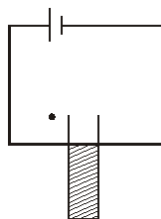


Space for Rough Work

6. An uncharged capacitor of capacitance C is connected in given circuit diagram, switch S is closed at $t = 0$. If current in branch BC as function of time is given by $I = I_0(\text{amp}) e^{-\frac{t}{\tau (\text{in } \mu\text{s})}}$ then find the numerical value of $I_0\tau$.



7. A dielectric slab of dielectric constant k , mass m , thickness d and area $L \times L$ is hanging vertically in equilibrium under the influence of gravity and electrostatic pull of a capacitor connected to a battery of voltage V . The capacitor has plates of area $L \times L$ and distance between plates is d . The capacitor is half filled by the dielectric. Suddenly a mass of m is attached to dielectric without any impulse on the system. The slab falls off in time t . Evaluate t (in sec). Take $k = 2$, $V = 4$ volts, $L = 80$ cm, $d = 0.1$ mm.



8. In an optical bench experiment to determine the focal length of the concave mirror, the least count of the bench scale (meter scale) is 0.1 cm. The object needle reading is 10.7 cm and the image needle reading is 5.7 cm. The mirror is positioned at 15.7 cm. The maximum possible percentage error in the focal length of the mirror with this set of readings is $\frac{10}{x}$. Find x .

Space for Rough Work

SECTION-IV : (Maximum Marks: 16)

- This section contains **TWO (02)** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **four** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **Column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- For each entry in **Column-I**, darken the options of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q) and (R), then chosen these three options. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

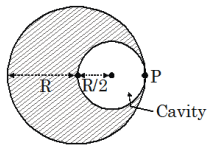
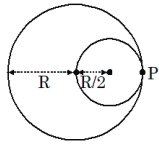
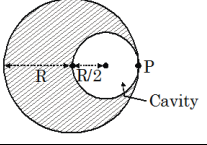
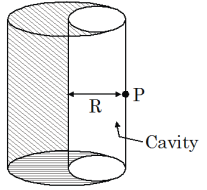
For each entry in **Column-I**

Full Marks : +2 If only the option(s) corresponding to all the correct match(es) is (are) chosen

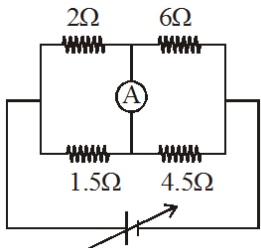
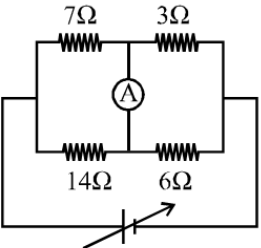
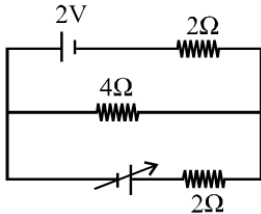
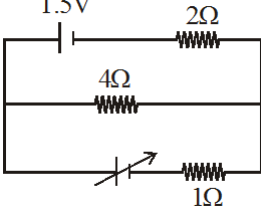
Zero Marks : 0 If none of the options is chosen

Negative Marks : -1 In all other cases

1. Column-I contains four charged systems of uniform charge density. Column-II contains potential at point P due to system in Column-I. Match Column-I with Column-II and select the correct answer using the code given below the column.

	Column-I		Column-II
(A)	Sphere of radius R having a cavity of radius R/2 with charge density ρ. 	(P)	$\frac{3\sigma R}{2\epsilon_0}$
(B)	Spherical shell of radius R having another spherical shell for radius R/2, both having charge density σ on their surface 	(Q)	$\frac{\sigma R}{2\pi\epsilon_0}$
(C)	Disc of radius R having a cut disc of radius R/2 with surface charge density σ 	(R)	$\frac{\rho R^2}{4\epsilon_0}$
(D)	Solid large cylinder of radius R having volume charge density ρ. Consider the potential at the axis of given cylinder as $\frac{\rho R^2}{2\epsilon_0}$. 	(S)	$\frac{3\rho R^2}{4\epsilon_0}$
		(T)	$\frac{\sigma R}{4\epsilon_0}$

2. In column-I, we have certain situations where a battery of variable voltage other than zero is connected in a circuit. Column-II has comments on current in element A on variation of voltage:-

	Column-I		Column-II
(A)	 Ammeter is ideal	(P)	On increasing V (from a certain value to another certain value), i may/will increase in magnitude
(B)	 Ammeter is ideal	(Q)	On increasing V (from a certain value to another certain value), i may/will decrease in magnitude.
(C)	 $A \rightarrow 4\Omega$ resistor	(R)	On increasing V (from a certain value to another certain value), i may/will not change at all in magnitude
(D)	 $A \rightarrow 4\Omega$ resistor	(S)	On reversing V (keeping same value), i may/will be zero
		(T)	On reversing V (keeping same value), i may/will increase in magnitude

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Mixture(s) showing positive deviation from Raoult's law at 35°C is (are)

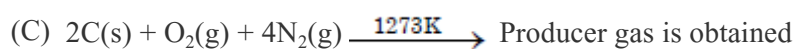
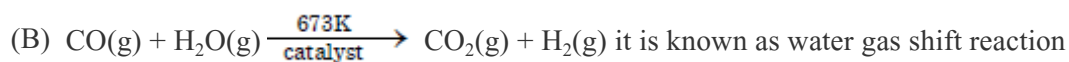
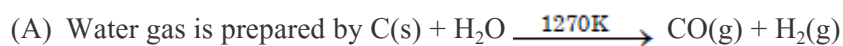
- (A) carbon tetrachloride + methanol
- (B) carbon disulphide + acetone
- (C) benzene + toluene
- (D) phenol + aniline

Space for Rough Work

2. The **CORRECT** statement(s) for cubic close packed (ccp) three dimensional structure is (are)

- (A) The number of the nearest neighbours of an atom present in the topmost layer is 12
- (B) The efficiency of atom packing is 74%
- (C) The number of octahedral and tetrahedral voids per atom are 1 and 2, respectively
- (D) The unit cell edge length is $2\sqrt{2}$ times the radius of the atom

3. Choose correct statement/s :-



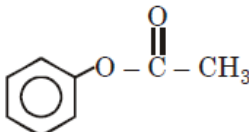
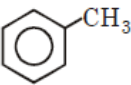
Space for Rough Work

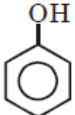
4. Which of the following statement is/are **CORRECT**?
- (A) The superoxide ion (i.e., O_2^-) is stable only in presence of larger cations such as K, Rb, Cs.
- (B) Na, K, Rb, Cs are normally kept in kerosene oil.
- (C) All the alkali metal hydrides are ionic solids with high melting points.
- (D) The highly concentrated solution of alkali metals in liquid ammonia are strong paramagnetic in nature.
5. Choose the **INCORRECT** option(s) for the following statements.
- (A) Spring reaction is used for the preparation of sodium thiosulphite.
- (B) $SO_2 + \text{Carbon} \xrightarrow{\Delta} A + B$, where SO_2 acts as reducing agent.
- (C) H_2SO_4 is better dehydrating agent than P_2O_5 .
- (D) $SO_2(g) + \frac{1}{2} O_2(g) \longrightarrow SO_3(g) \quad \Delta H = (+)ve$

Space for Rough Work

6. CORRECT statement(s) in the following?

(A) Rate of sulphonation of C_6H_6 and C_6D_6 are equal.

(B) Rate of halogenation of  is faster than 

(C)  on reaction with Br_2 / H_2O forms tribromoderivative.

(D)  on reaction with Cl_2 at $500^\circ C$ will give electrophilic aromatic substitution reaction.

7. Following statement(s) is/are true for D-glucose :

(A) It shows Schiff's reagent test

(B) It is C-2 epimer of D-mannose

(C) Oxidise with Br_2 / H_2O to form saccharic acid

(D) Shows mutarotation.

8. Optically active amino acid is :

(A) Glycine

(B) Alanine

(C) Valine

(D) Phenyl alanine

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 8)

- This section contains **FOUR (04)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. The osmotic pressure of an aq. urea solution 16.628 kPa at 27°C. The concentration of solution in %(w/v) is. (Use $R = 8.314 \text{ J/mole-K}$)
 2. Consider an ionic solid MX with NaCl structure. Construct a new structure (Z) whose unit cell is constructed from the unit cell of MX following the sequential instructions given below. Neglect the charge balance.
 - (i) Remove all the anions (X) except the central one
 - (ii) Replace all the face centered cations (M) by anions (X)
 - (iii) Remove all the corner cations (M)
 - (iv) Replace the central anion (X) with cation (M)

The value of $\left(\frac{\text{number of anions}}{\text{number of cations}} \right)$ in Z is ____.
 3. PCl_5 reacts with P_4O_{10} to give a single product. Calculate weight (in Kg) of PCl_5 required to react with one mole of P_4O_{10} in the balanced reaction (Given atomic weight of P = 31 g/mol & atomic weight of Cl = 35.5 g/mol)
 4. Number of oxygen atoms present in 1 molecule of $\beta\text{-D-2-deoxyribose}$
-

Space for Rough Work

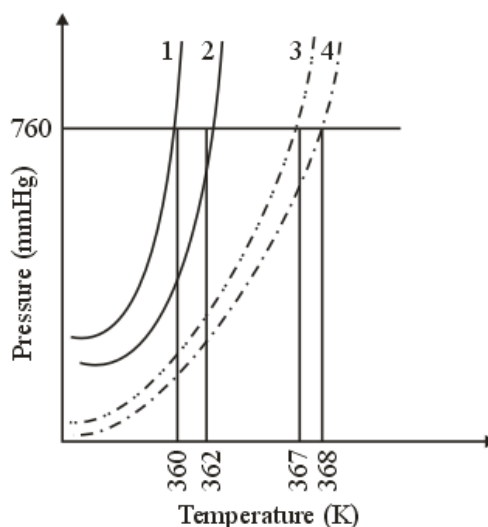
SECTION-II (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

5. The plot given below shows P-T curves (where P is the pressure and T is the temperature) for two solvents X and Y and isomolal solutions of NaCl in these solvents. NaCl completely dissociates in both the solvents.

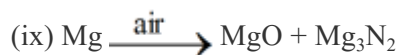
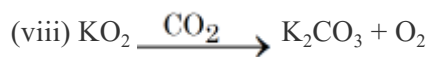
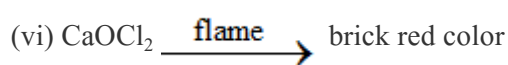
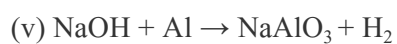
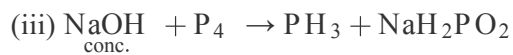
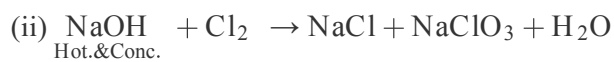
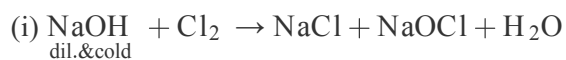


1. solvent X
2. solution of NaCl in solvent X
3. solvent Y
4. solution of NaCl in solvent Y

On addition of equal number of moles a non-volatile solute S in equal amount (in kg) of these solvents, the elevation of boiling point of solvent X is three times that of solvent Y. Solute S is known to undergo dimerization in these solvents. If the degree of dimerization is 0.7 in solvent Y, the degree of dimerization in solvent X is ____.

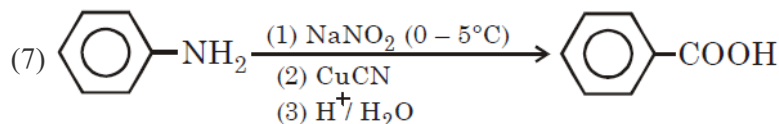
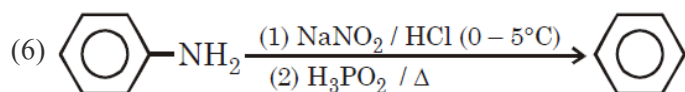
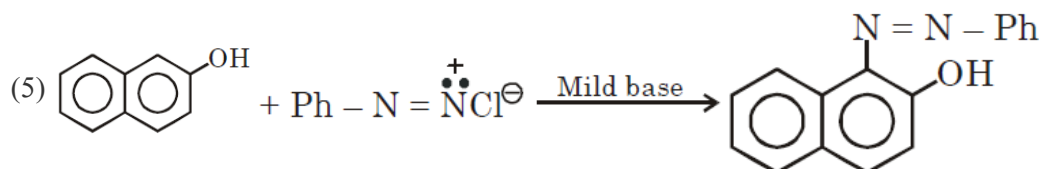
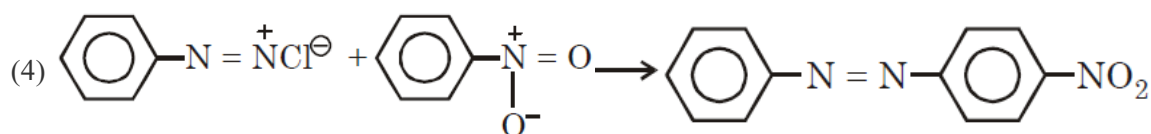
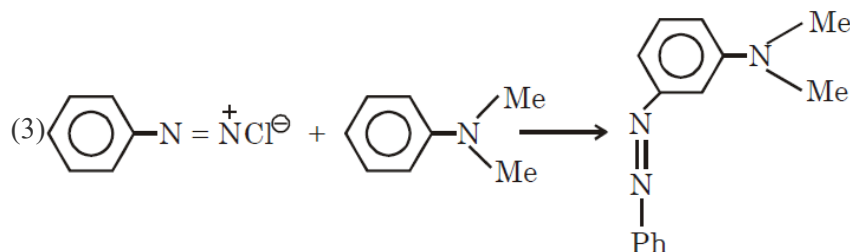
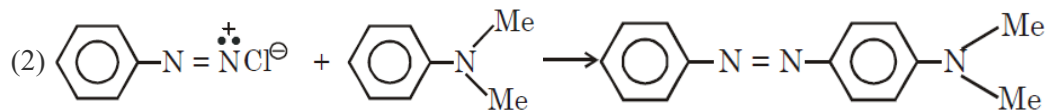
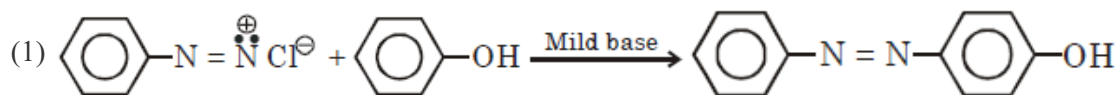
Space for Rough Work

6. Out of following how many reaction are correct.



Space for Rough Work

7. Number of reaction(s) which are correctly represented as major product?



8. Count number of polymers which are condensation polymers as well as co-polymers :

- (1) Polythene (2) Nylon-6, 6 (3) Nylon-6, 10 (4) Dacron
 (5) Bakelite (6) Melamine formaldehyde resin (7) Polyisoprene
 (8) Styrene-butadiene rubber (Buna-S) (9) PAN (10) Teflon

Space for Rough Work

SECTION-IV : (Maximum Marks: 16)

- This section contains **TWO (02)** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **four** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **Column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- For each entry in **Column-I**, darken the options of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q) and (R), then chosen these three options. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the option(s) corresponding to all the correct match(es) is (are) chosen

Zero Marks : 0 If none of the options is chosen

Negative Marks : -1 In all other cases

1.

Column-I		Column-II	
(A)	Cubic	(P)	Cell parameter $a = b = c$
(B)	Triclinic	(Q)	Cell parameter $a = b = g$
(C)	Hexagonal	(R)	Only one braivais lattic is possible
(D)	Orthorhombic	(S)	Four bravais lattices are possible
		(T)	Type of crystal system

2.

Column-I		Column-II	
(A)	$\text{Mg}_3\text{N}_2 + \text{H}_2\text{O} \longrightarrow$	(P)	Atleast one of the product is water soluble but all products are not water soluble
(B)	$\text{S} + \text{NaOH (conc.)} \longrightarrow$	(Q)	Disproportionation
(C)	$\text{CuSO}_4 \text{ sol.} + \text{PH}_3(\text{g}) \longrightarrow$	(R)	Non redox hydrolysis
(D)	$\text{Pb}_3\text{O}_4 + \text{dil. HNO}_3 \xrightarrow{\text{room temp.}}$	(S)	All products are water soluble
		(T)	Amphoteric compound is obtained

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors each of magnitude $\sqrt{3}$ and the angle between each pair of them is $\frac{\pi}{3}$.

If $\vec{b} \times \vec{c} + \vec{c} \times \vec{a} = x\vec{a} + y\vec{b} + z\vec{c}$ where x, y, z are scalars, then

(A) volume of parallelepiped formed by $\vec{a}, \vec{b}, \vec{c}$ is $\frac{3\sqrt{3}}{\sqrt{2}}$

(B) the value of x is $\frac{\sqrt{3}}{\sqrt{2}}$

(C) $x^2 + y^2 + z^2 \leq \sqrt{21}$

(D) Number of integral co-ordinates (x, y, z) satisfying $x^2 + y^2 + z^2 \leq 12$ is 125

Space for Rough Work

2. If $f(x) = \begin{cases} x^3 (\operatorname{sgn}[x] + \{x\}) & ; 0 \leq x < 2 \\ \cos x + |x - 2| & ; 2 \leq x < \pi \end{cases}$, then-

where $[\cdot]$, $\{\cdot\}$ and $\operatorname{sgn}(\cdot)$ represents the greatest integer, the fractional part and signum function respectively-

- (A) $f(x)$ is differentiable at $x = 1$
- (B) $f(x)$ is continuous but non-differentiable at $x = 1$
- (C) $f(x)$ is discontinuous at $x = 2$
- (D) $f(x)$ is non-differentiable at $x = 2$
3. $\vec{a}$ and $\vec{b}$ are non-collinear vectors such that $|\vec{a}| = 2$, $\vec{a} \cdot \vec{b} = 1$ and the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{3}$. If $\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$ satisfying $\vec{r} \cdot \vec{a} = 2$, $\vec{r} \cdot \vec{b} = 8$, $(\vec{r} + 5\vec{a} - 7\vec{b}) \cdot (\vec{a} \times \vec{b}) = 4\sqrt{3}$ then which of the following is true ?
- (A) $x + y + z = 6$
- (B) $x + y + z = 10$
- (C) $x + y + \sqrt{3}z = 12$
- (D) $x + y + \sqrt{3}z = 14$

Space for Rough Work

4. Consider the planes P_1, P_2 and points $A(2, 5, 1)$ and $B(1, 3, 0)$ where

$$P_1: x + y + z = 3$$

$$P_2: 2x + y - z = 4$$

then identify the correct statement(s)

(A) shortest distance between line AB and points common to P_1, P_2 is equal to $\frac{1}{5\sqrt{3}}$ units

(B) projection of segment AB on P_1 is equal to $\frac{\sqrt{6}}{3}$ units

(C) projection of segment AB on P_1 is equal to $2\sqrt{2}$

(D) area enclosed by locus of all points on P_1 which are at a distance of 3 units from B is equal to $\frac{26\pi}{3}$ sq. units

5. Let $f(x) = \begin{cases} e^{\tan^{-1}(x-1)} + (x-1)^{2017} \sin^2 \frac{1}{(x-1)} & \text{if } x \neq 1 \\ 1 & \text{if } x = 1 \end{cases}$ and $\ell = \lim_{n \rightarrow \infty} \left(n \sum_{k=1}^{10} f\left(1 + \frac{k}{n}\right) - n \sum_{k=1}^4 k \right)$

(A) $f(x)$ is continuous but non-derivable at $x = 1$

(B) $f(x)$ is differentiable at $x = 1$

(C) $55 f'(1) - \ell = 0$

(D) ℓ is divisible by 11.

Space for Rough Work

6. Consider $f : [0, \infty) \rightarrow [0, \infty)$; such that $f(x)$ is bijective and satisfies $f(x)e^{f(x)} = x$. Which of the following is/are true
- (A) The greatest integer less than or equal to the value of $f(f(e^{e+1}))$, is 1
- (B) The value of $\lim_{x \rightarrow \infty} \frac{f(x)}{\ln x}$, is 1
- (C) The greatest integer less than or equal to the value of $f(f(e^{e+1}))$, is 2
- (D) The value of $\lim_{x \rightarrow \infty} \frac{f(x)}{\ln x}$, is 3
7. If $f(x) = \lim_{n \rightarrow \infty} \frac{[1 - \cos(1 - \tan(\frac{\pi}{4} - x))] (x+1)^n + \lambda \sin((n - \sqrt{n^2 - 8n})x)}{x^2(x+1)^n + x}$, $x \neq 0$ is continuous at $x = 0$ then which of the following is/are true ($f(0) + 2\lambda$)
- (A) $f(0) = 2$ (B) $\lambda = 1$ (C) $f(0) = 3$ (D) $\lambda = 1/2$
8. If $y = x^{n-1} \ln x$, satisfies $x^2 y_2 + f(n) x y_1 + g(n) y = 0$. where $f(n)$ and $g(n)$ are function of n ($n \in \mathbb{N}$) then (y_1 and y_2 denote first order and second order derivative wrt. x) :
- (A) $f(3) - g(3) = 7$ (B) $f(4) + g(4) = 4$
- (C) $f(5) \times g(5) = -112$ (D) None of these

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 8)

- This section contains **FOUR (04)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If $f(x) = \begin{cases} (x+3)|x^2+5x+6| & ; x < 0 \\ (x-1)[x] & ; 0 \leq x \leq 2 \\ (x-1)(x-3)^{1/3} & ; x > 2 \end{cases}$

Then number of values of x where $f(x)$ is continuous but not differentiable is/are (where $[\cdot]$ represent G.I.F.)

2. If the range of real-valued function $f(x) = \sqrt{\cos^{-1}(2x) + 2} + \sqrt{1 - \cos^{-1}(2x)}$ is $[\sqrt{a}, \sqrt{b} + 1]$ (where a and b are positive integers) then $\frac{(a+b)}{20}$ is equal to

3. If the number of positive integral value(s) of ' x ' such that the interval $\left[\frac{x}{10}, \frac{3x}{10}\right]$ contains exactly 3 integers is λ , then $\frac{\lambda}{5}$ equals

4. Let $f(x) = \frac{a^x}{a^x + \sqrt{a}}$, $a > 1$ and $\int_0^2 \left(f\left(\frac{1}{100}\right) + f\left(\frac{2}{100}\right) + f\left(\frac{3}{100}\right) + \dots + f\left(\frac{99}{100}\right) \right) \cdot dx = k \times 11$
then $\frac{k}{27}$ equals

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

5. Let $f(x) = \cos^{-1} \left(\frac{1-x^2-2x}{(x+1)^2} \right)$ & $g(x) = 1-x$. If the number of solutions for $f(x) = g(x)$ when $x > 0$ is λ , then $\frac{\lambda}{4}$ equals
6. Let $\vec{a}$, $\vec{b}$, $\vec{c}$ are three vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $|\vec{c}| = 5$, then the value of $\frac{|\vec{a} + \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} + \vec{a}|^2}{19}$ is equal to :
7. Let $f(x) = ax^3 + bx^2 + cx + d$, satisfying $\lim_{x \rightarrow 0} \left(3 + \frac{f(x)}{x^2} \right)^{\frac{1}{x}} = e^2$, then $(a+c)$ equals
8. Given $f(x).f(y) + f\left(\frac{a}{x}\right) \cdot f\left(\frac{a}{y}\right) = 2f(xy) \forall x, y \in \mathbb{R}^+$, where a is also a positive real number such that $f(a) = 1$, then $\left(f(1) - (f(2))^2 + \sqrt{f(3)} \right)$ is equal to

Space for Rough Work

SECTION-IV : (Maximum Marks: 16)

- This section contains **TWO (02)** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **four** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **Column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- For each entry in **Column-I**, darken the options of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q) and (R), then chosen these three options. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the option(s) corresponding to all the correct match(es) is (are) chosen

Zero Marks : 0 If none of the options is chosen

Negative Marks : -1 In all other cases

1.	Column-I	Column-II	
(A)	Let $\tan(2\pi \sin \theta) = \cot(2\pi \cos \theta)$, where $\theta \in \mathbb{R}$ and $f(x) = (\sin \theta + \cos \theta)^x$ then the value of $\lim_{x \rightarrow \infty} \left[\frac{2}{f(x)} \right] =$ where [G.I.F.]	(P)	0
(B)	If $\tan^2 \alpha + \cot^2 \alpha = \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 8x + 3} - \sqrt{x^2 + 4x + 2} \right)$ where $\alpha \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$ then the possible value of $\frac{1}{2} (\tan \alpha + \cot \alpha)$	(Q)	1
(C)	If $f(x) = [2 + 5 n \sin x]$ where $n \in \mathbb{I}$, $x \in [0, \pi]$ has exactly 19 points of non differentiability then the possible value of $ n $ is	(R)	4
(D)	If $n \in \mathbb{N}$ and the set of equations $\cos^{-1} x + (\sin^{-1} y)^2 = \frac{n\pi^2}{4}$ and $(\sin^{-1} y)^2 - (\cos^{-1} x) = \frac{\pi^2}{16}$ is consistent then 'n' can be equal to	(S)	2
		(T)	5

Space for Rough Work

2.

Column-I		Column-II	
(A)	If $\vec{a}$ & $\vec{b}$ are two non collinear unit vector a. $\vec{b} \neq 0$ $\vec{a} \times (\underbrace{\vec{a} \times (\vec{a} \times (\vec{a} \times \dots \times (\vec{a} \times (\vec{a} \times \vec{b}))}_{2017 \text{ times}})) = m(\vec{a} \times \vec{b})$ then m equals	(P)	2
(B)	Let OABC is a regular tetrahedron and P is any point in space. If edge length of tetrahedron is 1 unit, the least value of $2(PA^2 + PB^2 + PC^2 + PO^2)$ is	(Q)	1
(C)	If D, E, F be three point on BC, CA, AB respectively of a ΔABC such that the line AD, BE, CF are concurrent then the value of $\frac{BD}{CD} \cdot \frac{CE}{AE} \cdot \frac{AF}{BF}$ is	(R)	3
(D)	Let $\vec{a}$, $\vec{b}$, $\vec{c}$ are three unit vectors inclined with each other by an angle of $\frac{\pi}{3}$. If $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$; $\vec{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$; $\vec{c} = c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k}$ and A be the matrix of order 3×3 such that $\text{adj}(\text{adj} A) = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$ then $4 A ^8$ equals	(S)	4
		(T)	5

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work